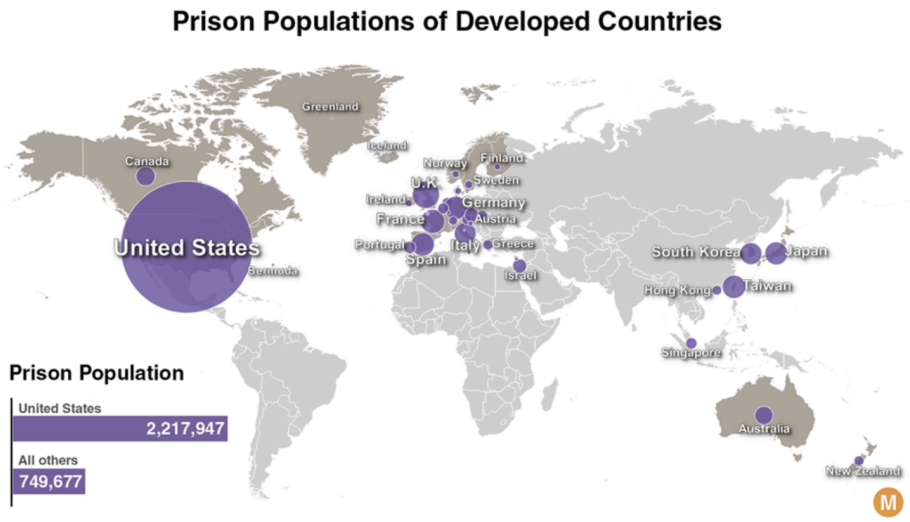
Visualisation 1

Website: https://metrocosm.com/prison-population-map/ **(circa 2016 - link no longer working)**

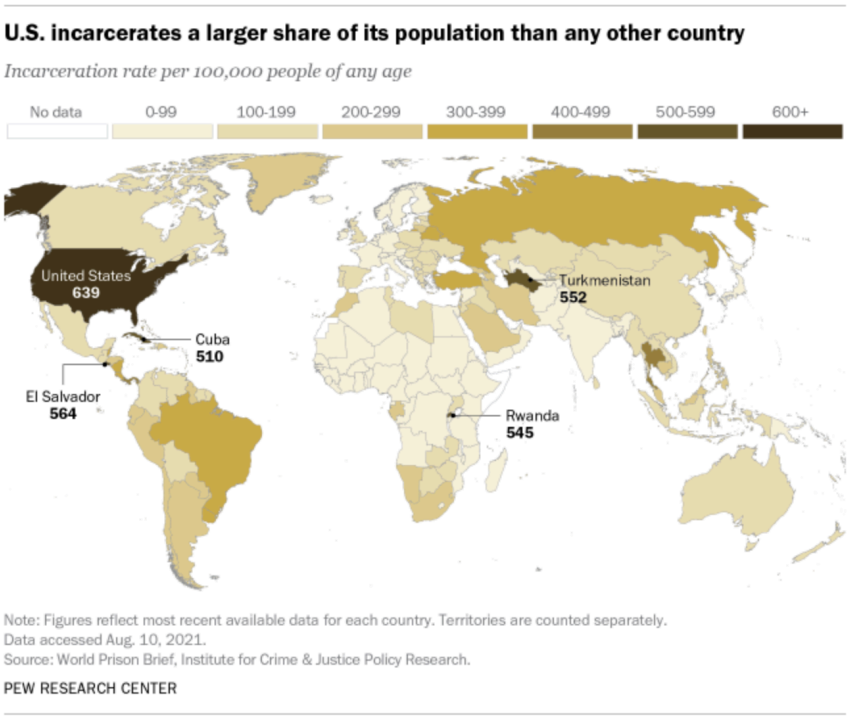


Visualisation 1

- **Purpose:** To emphasize the sheer volume of the United States' prison population compared to other developed nations.
- **Chart Type:** A proportional symbol map (bubble map) paired with a stacked bar chart.
- **Data Encoding:**
    - **Marks:** Bubbles (circles) for the map; lines/rectangles for the bar chart.
    - **Channels:** The map uses 2D spatial position for geographic location and bubble area (size) to encode the magnitude of the prison population. The bar chart uses line length to encode magnitude.
- **Interactive Features:** None are explicitly visible in the static screenshot, though web-based maps of this style often feature hover tooltips to reveal exact numbers.
- **Messaging Features:** It utilizes a massive, overlapping purple bubble for the U.S. that visually swallows North America, creating immediate shock value. The accompanying bar chart explicitly groups the U.S. against "All others" to reinforce this disparity.
- **Advantages:** Creates an immediate, visceral impact. It is excellent for highlighting massive outliers.
- **Disadvantages:** Humans are poor at accurately comparing the areas of circles. Furthermore, the overlapping bubbles in Europe obscure geographic details and make it impossible to differentiate specific countries.

Visualisation 2

Website: https://www.pewresearch.org/short-reads/2021/08/16/americas-incarceration-rate-lowest-since-1995



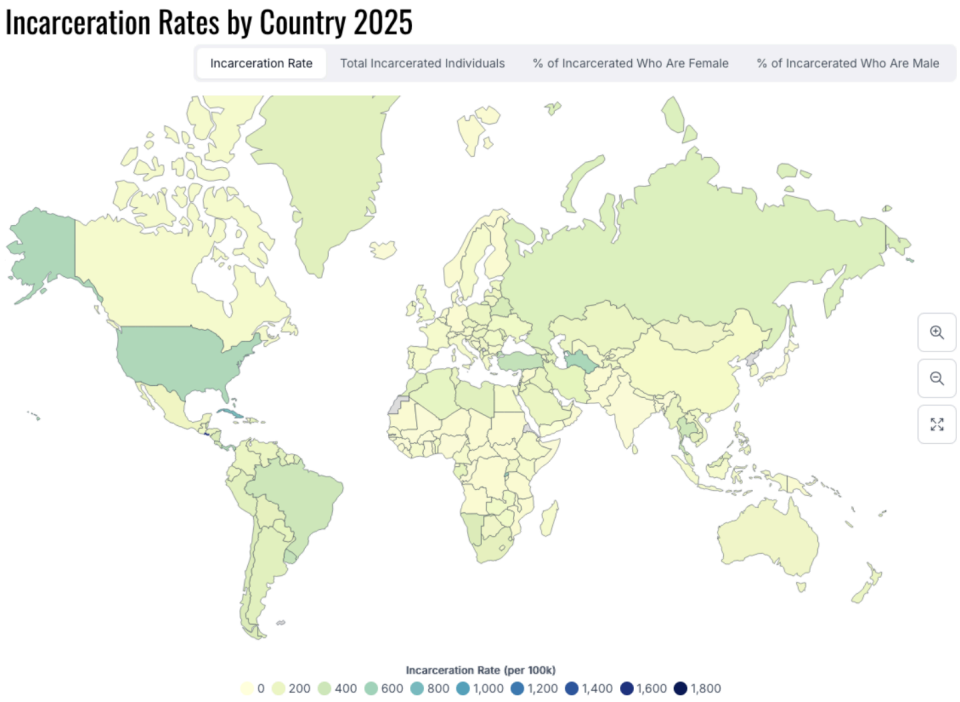
Visualisation 2

- **Purpose:** To demonstrate that the U.S. has the highest per capita incarceration rate globally.
- **Chart Type:** Choropleth map.
- **Data Encoding:**
    - **Marks:** Geographic areas (country borders).
    - **Channels:** Color saturation/luminance based on a sequential color scale (ranging from light beige to dark brown) to encode incarceration rate bins (e.g., 0-99, 100-199).
- **Interactive Features:** None visible.
- **Messaging Features:** Uses a bold, direct headline ("U.S. incarcerates a larger share..."). It also features specific data callouts with exact numbers for selected countries (e.g., U.S. 639, El Salvador 564) to provide immediate context without needing to check the legend.
- **Advantages:** Good for showing geographic trends and broad regional distributions of data.
- **Disadvantages:** Large countries visually dominate the map, giving disproportionate visual weight to low-density areas (like Russia or Canada), while small countries with

high rates are easily missed. Binning data into ranges can obscure significant differences between countries within the same bin.

Visualisation 3

Website: https://worldpopulationreview.com/country-rankings/incarceration-rates-by-country



Visualisation 3

- **Purpose:** To provide a comprehensive, exploratory overview of global incarceration statistics.
- **Chart Type:** Choropleth map.
- **Data Encoding:**
    - **Marks:** Geographic areas.
    - **Channels:** Color hue and saturation (ranging from yellow to dark blue) to encode exact incarceration rates per 100k.
- **Interactive Features:** Highly interactive. It includes zoom in (+), zoom out (-), and full-screen controls on the right. Crucially, it features clickable tabs at the top ("Incarceration Rate", "Total Incarcerated Individuals", "% of Incarcerated Who Are Female", "% of Incarcerated Who Are Male") that allow the user to change the dataset being visualized entirely.
- **Messaging Features:** Unlike the previous charts, this one is neutral. It provides a clean dashboard for exploration rather than forcing a specific narrative.
- **Advantages:** Excellent for user-driven data exploration and finding precise information across multiple metrics.

- **Disadvantages:** It lacks a strong, immediate takeaway. Users must actively explore to find insights.

Visualisation 4

Reference: https://www.prisonpolicy.org/global/2024.html

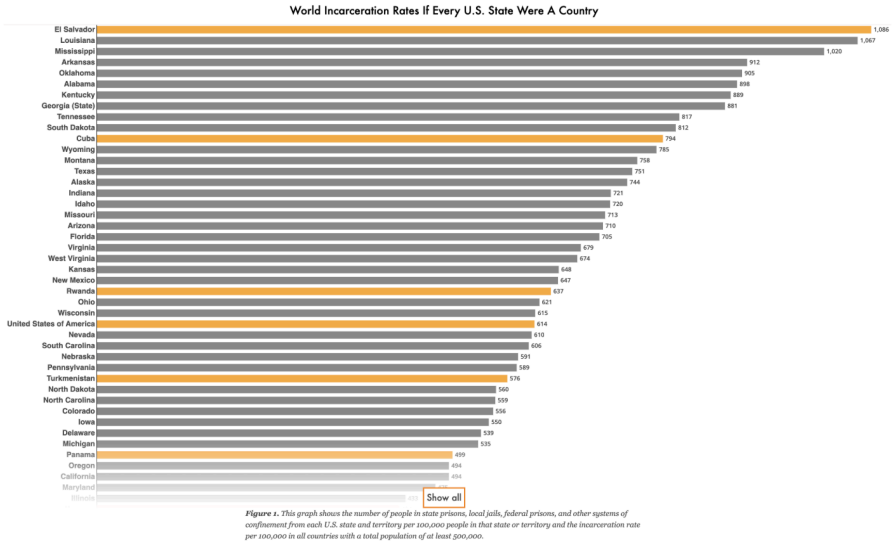


Figure 1. This graph shows the number of people in state prisons, local jails, federal prisons, and other systems of confinement from each U.S. state and territory per 100,000 people in that state or territory and the incarceration rate per 100,000 in all countries with a total population of at least 500,000.

Visualisation 4

- **Purpose:** To rank individual U.S. states against independent countries globally to show the extreme punitive nature of localized U.S. policies.
- **Chart Type:** Horizontal bar chart.
- **Data Encoding:**
    - **Marks:** Lines/bars.
    - **Channels:** Length encodes magnitude (incarceration rate). Vertical position encodes categorical ranking (highest to lowest). Color hue categorizes the data (grey for U.S. states, orange/gold for independent countries and the U.S. national average).
- **Interactive Features:** A "Show all" button at the bottom indicates the list is truncated to save space and can be expanded by the user.
- **Messaging Features:** The specific use of color highlighting forces the viewer to realize how many individual U.S. states (grey) sit above strict authoritarian nations globally.
- **Advantages:** Humans are highly accurate at comparing line lengths. Sorting by magnitude makes ranking instantly readable.
- **Disadvantages:** Can become visually overwhelming or dense with too many categories, requiring small text and scrolling.

Visualisation 5

Reference: https://www.prisonpolicy.org/global/2024.html (scroll down)

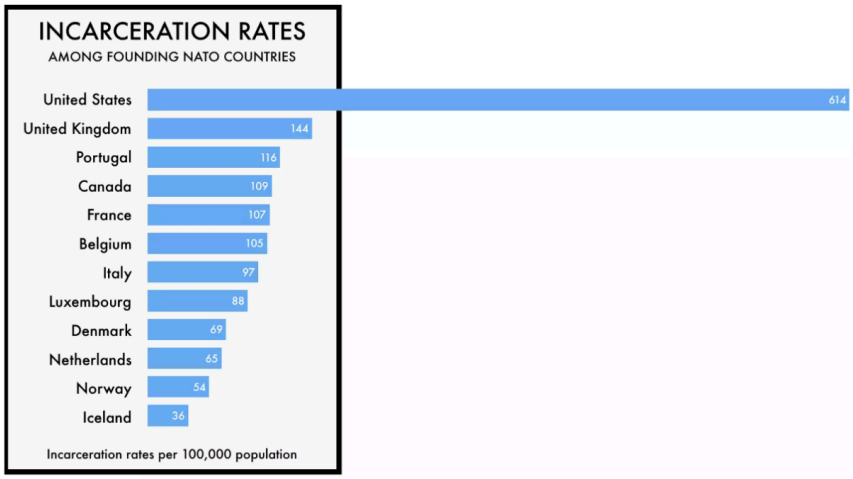


Figure 2. *Compare another state:* U.S.

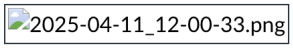
Visualisation 5

- **Purpose:** To highlight the incarceration disparity between the U.S. and its closest geopolitical/economic peers (founding NATO countries).
- **Chart Type:** Horizontal bar chart.
- **Data Encoding:**
    - **Marks:** Lines/bars.
    - **Channels:** Length encodes magnitude.
- **Interactive Features:** A dropdown menu at the bottom labeled "Compare another state: [U.S. v]" allows the user to dynamically swap the U.S. national average with specific U.S. states to see how they stack up against NATO peers.
- **Messaging Features:** The strict filtering of the data (apples-to-apples comparison among similar wealthy democracies) isolates the U.S. as an undeniable extreme outlier.
- **Advantages:** Highly focused, readable, and delivers a singular, undeniable message without clutter.
- **Disadvantages:** Limited scope; it omits the rest of the world, which might leave users wondering about broader global context.

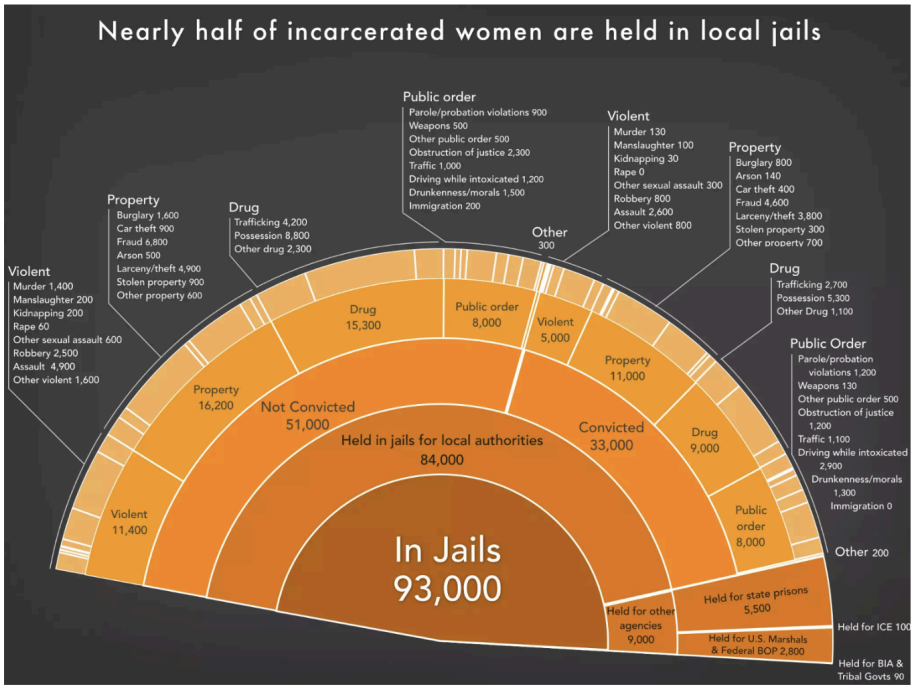
Visualisation 6

Reference: https://www.prisonpolicy.org/reports/pie2024women.html

If you have time...



and...



Visualisation 6

- **Purpose:** To break down the demographics, legal status, and offenses of incarcerated women, specifically highlighting that nearly half are in local jails.
- **Chart Type:** Half-sunburst chart (a hierarchical nested donut chart).
- **Data Encoding:**
    - **Marks:** Arcs/areas.
    - **Channels:** Arc length/angle encodes magnitude (population count). Color hue differentiates broad categories. Radial position (inner to outer rings) encodes hierarchical levels (Facility -> Conviction Status -> Offense).
- **Interactive Features:** None apparent in this static capture.
- **Messaging Features:** The massive, visually dominant orange slice for "Not Convicted" perfectly supports the headline, drawing the eye immediately to the systemic issue of pre-trial detention.
- **Advantages:** Excellent for showing part-to-whole relationships across multiple nested sub-categories in a single view.

- **Disadvantages:** Humans struggle to accurately estimate and compare angles and arc lengths, making it difficult to compare specific slices (e.g., comparing "Violent 11,400" to "Property 16,200") without reading the text labels